

Topic 10: Hypothesis Testing

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The Problem of Hypothesis Testing

- A **statistical hypothesis** is an assertion or conjecture about the probability distribution of one or more random variables.
- A **test of a statistical hypothesis** is a rule or procedure for deciding whether to reject that assertion.
- Suppose we have a sample $x = (x_1, \dots, x_n)$ from a density f . We have two hypotheses about f . On the basis of our sample, one of the hypotheses is **accepted** and the other is **rejected**.
- The two hypotheses have different status:
 - the **null hypothesis**, H_0 , is the hypothesis under test. It is the conservative hypothesis, not to be rejected unless the evidence is clear
 - the **alternative hypothesis** H_1 specifies the kind of departure from the null hypothesis that is of interest to us
- A hypothesis is **simple** if it completely specifies the probability distribution, else it is **composite**.

Examples: (i) Income is log-normally distributed with known variance but unknown mean. $H_0 : \mu \geq 8,000$ rupees per month, $H_1 : \mu < 8,000$ (ii) We would like to know whether parents are more likely to have boys than girls. Denoting with the probability of a boy child being denoted by the Bernoulli parameter p . $H_0 : p = \frac{1}{2}$ and $H_1 : p > \frac{1}{2}$

Statistical tests

- Before deciding whether or not to accept H_0 , we observe a random sample. Denote by S , the set of all possible outcomes X of the random sample.
- A **test procedure** partitions the set S into two subsets, one containing the values that will lead to the acceptance of H_0 and the other containing values which lead its rejection.
- A **statistical test** is defined by the **critical region R** which is the subset of S for which H_0 is rejected. The complement of this region must therefore contain all outcomes for which H_0 is not rejected.
- Tests are based on values taken by a **test statistic** (the same mean, the sample variance or functions of these). In this case, the critical region R , is a subset of values of the test statistic for which H_0 is rejected. The **critical values** of a test statistic are the bounds of R .
- When arriving at a decision based on a sample and a test, we may make two types of errors:
 - H_0 may be rejected when it is true- a **Type I** error
 - H_0 may be accepted when it is false- a **Type II** error
- Since a test is any rule which we use to make a decision and we have to think of rules that help us make better decisions (in terms of these errors).
- We will first discuss how to evaluate a given test using its **power function**, and then consider situations in which we have **optimal tests** and derive these.

The power function

- One way to characterize a test is to specify, for each value of $\theta \in \Omega$, the probability $\pi(\theta)$ that the test procedure will lead to the rejection of H_0 . The function $\pi(\theta)$ is called the **power function** of the test:

$$\pi(\theta) = \Pr(X \in R) \text{ for } \theta \in \Omega$$

If we are using a test statistic T , $\pi(\theta) = \Pr(T \in R)$ for $\theta \in \Omega$

- Since the **power function** of a test specifies the probability of rejecting H_0 as a function of the real parameter value, we can evaluate our test by asking how often it leads to mistakes. What is the power function of an **ideal test** ? Think of examples when such a test exists.
- It is common to specify an upper bound α_0 on $\pi(\theta)$ for every value $\theta \in \Omega_0$. This bound α_0 is the **level of significance** of the test.
- The **size of a test**, α is the maximum probability, among all values of $\theta \in \Omega_0$ of making an incorrect decision:

$$\alpha = \sup_{\theta \in \Omega_0} \pi(\theta)$$

- Given a level of significance α_0 , only tests for which $\alpha \leq \alpha_0$ are admissible.

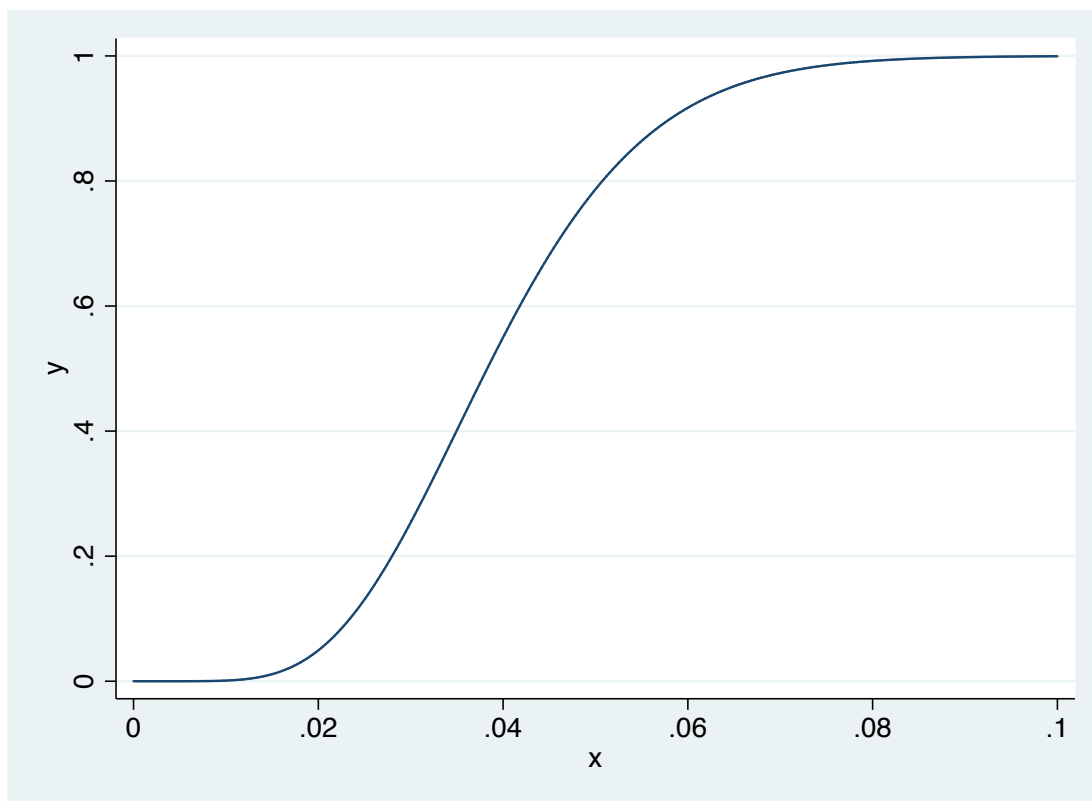
Example 1: Binomial distribution

- We want to test a hypothesis about the number of defective bolts in a shipment. The probability of a bolt being defective is assumed to follow a Bernoulli distribution.
- Consider $H_0 : p \leq .02$, $H_1 : p > .02$.
- We have a sample of 200 items and our test takes the form of rejecting the null hypothesis if we find more than a certain number of items to be defective. We want to find a test for which $\alpha_0 = .05$.
- The number of defective items $X \sim \text{Bin}(n, p)$ and $P(x)$ is increasing in p for all $x > np$. We can therefore restrict our attention $p = .02$ when finding a test with significance level $\alpha_0 = 0.05$.
- It turns out that for $p = .02$, the probability that the number of defective items is greater than 7 is .049 (`display 1-binomial(200,7,.02)`).
- $R = \{x : x > 7\}$ is therefore the test we choose and its size is 0.049. With discrete distributions, the size will often be strictly smaller than α_0 . The size of tests which reject for more than 4, 5 and 6 defective pieces are .37. .21 and .11 respectively.

Example 1: power function

Let's graph the power function of this test:

twoway function 1-binomial(200,7,x), range(0 .1)



What happens to this power function as we increase or decrease the critical region? (say $R = \{x : x > 6\}$ or $R = \{x : x > 8\}$). Can you mark off the two types of errors for different values of p ?

Example 2 : Uniform distribution

A random sample is taken from a **uniform distribution** on $[0, \theta]$ and we would like to test

$$H_0 : 3 \leq \theta \leq 4 \quad \text{against} \quad H_1 : \theta < 3 \text{ or } \theta > 4$$

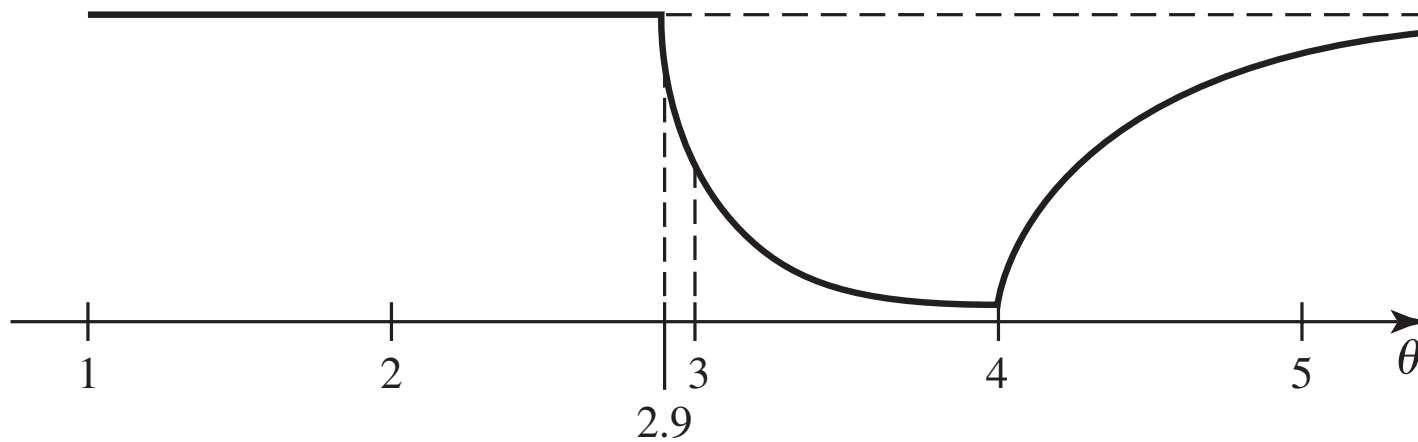
Our test procedure uses the M.L.E. of θ , $Y_n = \max(X_1, \dots, X_n)$ and rejects the null hypothesis whenever Y_n lies outside $[2.9, 4]$. What might be the rationale for this type of test?

The power function for this test is given by

$$\pi(\theta) = \Pr(Y_n < 2.9 | \theta) + \Pr(Y_n > 4 | \theta)$$

- What is the power of the test if $\theta < 2.9$?
- When θ takes values between 2.9 and 4, the probability that any sample value is less than 2.9 is given by $\frac{2.9}{\theta}$ and therefore $\Pr(Y_n < 2.9 | \theta) = (\frac{2.9}{\theta})^n$ and $\Pr(Y_n > 4 | \theta) = 0$. Therefore the power function $\pi(\theta) = (\frac{2.9}{\theta})^n$
- When $\theta > 4$, $\pi(\theta) = (\frac{2.9}{\theta})^n + 1 - (\frac{4}{\theta})^n$

The power graph..example 2



Example 3: Normal distribution

- $X \sim N(\mu, 100)$ and we are interested in testing $H_0 : \mu = 80$ against $H_1 : \mu > 80$.
- Let $\bar{x}$ denote the mean of a sample $n = 25$ from this distribution and suppose we use the critical region $R = \{(x_1, x_2, \dots, x_{25}) : \bar{x} > 83\}$.
- The power function is

$$\pi(\mu) = P(\bar{X} > 83) = P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{83 - \mu}{2}\right) = 1 - \Phi\left(\frac{83 - \mu}{2}\right)$$

- The size of this test is the probability of Type 1 error: $\alpha = 1 - \Phi\left(\frac{3}{2}\right) = .067 = \pi(80)$
- What are the values of $\pi(83)$, and $\pi(86)$?

$\pi(80)$ is given above, $\pi(83) = 0.5$, $\pi(86) = 1 - \Phi\left(-\frac{3}{2}\right) = \Phi\left(\frac{3}{2}\right) = .933$

`stata: display normal(1.5)`

- We can sketch the graph of the power function using the command
`stata: twoway function 1-normal((83-x)/2), range (70 90)`
- What is the p-value corresponding to $\bar{x} = 83.41$?

This is the **smallest level of significance, α_0** at which a given hypothesis would be rejected based on the observed outcome of X ?

In this case, $\Pr(\bar{X} \geq 83.41) = 1 - \Phi\left(\frac{3.41}{2}\right) = .044$.

Testing simple hypotheses

- Suppose that Ω_0 and Ω_1 contain only a single element each and our null and alternative hypotheses are given by

$$H_0 : \theta = \theta_0 \text{ and } H_1 : \theta = \theta_1$$

- Denote by $f_i(\mathbf{x})$ the joint density function or p.f. of the observations in our sample under H_i :

$$f_i(\mathbf{x}) = f(x_1|\theta_i)f(x_2|\theta_i)\dots f(x_n|\theta_i)$$

- Denote the type I and type II errors by $\alpha(\delta)$ and $\beta(\delta)$ respectively:

$$\alpha(\delta) = \Pr(\text{Rejecting } H_0 | \theta = \theta_0) \quad \text{and} \quad \beta(\delta) = \Pr(\text{Not Rejecting } H_0 | \theta = \theta_1)$$

- By always accepting H_0 , we achieve $\alpha(\delta) = 0$ but then $\beta(\delta) = 1$. The converse is true if we always reject H_0 .
- It turns out that we can find an optimal test which minimizes any linear combination of $\alpha(\delta)$ and $\beta(\delta)$.

Optimal tests for simple hypotheses

The test procedure δ that minimizes $a\alpha(\delta) + b\beta(\delta)$ for specified constants a and b :

THEOREM : Let δ^* denote a test procedure such that the hypothesis H_0 is accepted if $af_0(x) > bf_1(x)$ and H_1 is accepted if $af_0(x) < bf_1(x)$. Either H_0 or H_1 may be accepted if $af_0(x) = bf_1(x)$. Then for any other test procedure δ ,

$$a\alpha(\delta^*) + b\beta(\delta^*) \leq a\alpha(\delta) + b\beta(\delta)$$

So we reject whenever the likelihood ratio $\frac{f_1(x)}{f_0(x)} > \frac{a}{b}$. If we are minimizing the sum of errors, we would reject whenever $\frac{f_1(x)}{f_0(x)} > 1$.

Proof. (for discrete distributions)

$$a\alpha(\delta) + b\beta(\delta) = a \sum_{x \in R} f_0(x) + b \sum_{x \in R^c} f_1(x) = a \sum_{x \in R} f_0(x) + b \left[1 - \sum_{x \in R} f_1(x) \right] = b + \sum_{x \in R} [af_0(x) - bf_1(x)]$$

The desired function $a\alpha(\delta) + b\beta(\delta)$ will be minimized when the above summation is minimized. This will happen if the critical region includes only those points for which $af_0(x) - bf_1(x) < 0$. We therefore reject when the likelihood ratio exceeds $\frac{a}{b}$.

□

Minimizing $\beta(\delta)$, given α_0

If we fix a level of significance α_0 we want a test procedure that minimizes $\beta(\delta)$, the type II error subject to $\alpha \leq \alpha_0$.

The Neyman-Pearson Lemma : Let δ^* denote a test procedure such that, for some constant k , the hypothesis H_0 is accepted if $f_0(x) > kf_1(x)$ and H_1 is accepted if $f_0(x) < kf_1(x)$. Either H_0 or H_1 may be accepted if $f_0(x) = kf_1(x)$. If δ is any other test procedure such that $\alpha(\delta) \leq \alpha(\delta^*)$, then it follows that $\beta(\delta) \geq \beta(\delta^*)$. Furthermore if $\alpha(\delta) < \alpha(\delta^*)$ then $\beta(\delta) > \beta(\delta^*)$

This result implies that if we set a level of significance $\alpha_0 = .05$, we should try and find a value of k for which $\alpha(\delta^*) = .05$. This procedure will then have the minimum possible value of $\beta(\delta)$.

Proof. (for discrete distributions)

From the previous theorem we know that $\alpha(\delta^*) + k\beta(\delta^*) \leq \alpha(\delta) + k\beta(\delta)$. So if $\alpha(\delta) \leq \alpha(\delta^*)$, it follows that $\beta(\delta) \geq \beta(\delta^*)$

□

Neyman Pearson Lemma..example 1

Let $X_1 \dots X_n$ be a sample from a normal distribution with unknown mean and variance 1.

$$H_0 : \theta = 0 \text{ and } H_1 : \theta = 1$$

We will find a test procedure δ which keeps $\alpha = .05$ and minimizes β . We have

$$f_0(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2} \sum x_i^2} \text{ and } f_1(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2} \sum (x_i - 1)^2}$$

$$\frac{f_1(\mathbf{x})}{f_0(\mathbf{x})} = e^{n(\bar{x}_n - \frac{1}{2})}$$

The lemma tells us to use a procedure which rejects H_0 when the likelihood ratio is greater than a constant k . This condition can be re-written in terms of our sample mean $\bar{x}_n > \frac{1}{2} + \frac{1}{n} \log k = k'$. We want to find a value of k' for which $\Pr(\bar{X}_n > k' | \theta = 0) = .05$ or, alternatively, $\Pr(Z > k' \sqrt{n}) = .05$ (why?)

This gives us $k' \sqrt{n} = 1.645$ or $k' = \frac{1.645}{\sqrt{n}}$. Under this procedure, the type II error, $\beta(\delta)$ is given by

$$\beta(\delta^*) = \Pr(\bar{X}_n < \frac{1.645}{\sqrt{n}} | \theta = 1) = \Pr(Z < 1.645 - \sqrt{n})$$

For $n = 9$, we have $\beta(\delta^*) = 0.0877$ (`display normal(1.645-3)`)

If instead, we are interested in choosing δ_0 to minimize $2\alpha(\delta) + \beta(\delta)$, we choose $k' = \frac{1}{2} + \frac{1}{n} \log 2$, so with $n = 9$ our optimal procedure rejects H_0 when $\bar{x}_n > 0.577$. In this case, $\alpha(\delta_0) = 0.0417$ (`display 1-normal( (.577)*3)`) and $\beta(\delta_0) = 0.1022$ (`display normal( (.577-1)*3)`) and the minimized value of $2\alpha(\delta) + \beta(\delta)$ is 0.186

Neyman Pearson Lemma..example 2

Let $X_1 \dots X_n$ be a sample from a **Bernoulli distribution**

$$H_0 : p = 0.2 \text{ and } H_1 : p = 0.4$$

We will find a test procedure δ which keeps $\alpha = .05$ and minimizes β . let $Y = \sum X_i$ and y its realization. We have

$$f_0(x) = (0.2)^y (0.8)^{n-y} \text{ and } f_1(x) = (0.4)^y (0.6)^{n-y}$$

$$\frac{f_1(x)}{f_0(x)} = \left(\frac{3}{4}\right)^n \left(\frac{8}{3}\right)^y$$

The lemma tells us to use a procedure which rejects H_0 when the likelihood ratio is greater than a constant k . This condition can be re-written in terms of our sample mean $y > \frac{\log k + n \log \frac{4}{3}}{\log \frac{8}{3}} = k'$.

Now we would like to find k' such that $\Pr(Y > k' | p = 0.2) = .05$. We may not however be able to do this given that Y is discrete. If $n = 10$, we find that $\Pr(Y > 3 | p = 0.2) = .121$ and

$\Pr(Y > 4 | p = 0.2) = .038$, (**display 1-binomial(10,4,.2)**) so we can decide to set these probabilities as the values of $\alpha(\delta)$ and use the corresponding values for k' .